

This approach is very different from slamming a value-growth factor into a mean-variance optimizer. We start with the needs and characteristics of the asset owner (his *utility function*, in the language of an economist). We compare his bad outcomes with the bad times as defined by the value-growth factor. This requires an economic understanding, not just a statistical optimization procedure, of why these risk premiums exist. The appropriateness of a factor strategy depends on whether the investor can tolerate the factor’s bad times more readily than the average investor, or how he differs in behavior from the average investor.

3.5. SUMMARY

Factor investing compares how you feel about bad times to how the average investor feels. If you are average—and there is nothing wrong with being average—hold the market portfolio. If you differ from average, you tilt to dynamic factor risk exposures. You hold factors whose losses during bad times can be endured more easily by you relative to the typical investor. Or you might hold negative positions in factors—which is equivalent to buying insurance against factor losses.

4. Dynamic Factor Benchmarks

Dynamic factors imply nonmarket capitalization weighted holdings, which change through time. Unlike the CAPM, there is no unique way to construct an optimal dynamic factor portfolio.

4.1. MECHANICS

Consider three stocks, which we label “Growth,” “Neutral,” and “Value” with market weights 20%, 50%, and 30%, respectively.²⁸ We assume that these are the only three stocks traded, and we have given these labels on the basis of the stock prices relative to fundamental value. We construct a value-growth factor that is long the value stock and short the growth stock. Suppose the desired portfolio is 100% equities with a loading of 5% on the value-growth portfolio. The optimal benchmark is the market portfolio plus the value-growth:

	Market Portfolio	Value-Growth Factor	Benchmark Portfolio
Value	0.20	0.05	0.25
Neutral	0.50		0.50
Growth	0.30	−0.05	0.25
Sum Weights	1.00	0.00	1.00

²⁸ This is drawn from Section IIIB of Ang, Goetzmann, and Schaefer (2009).